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# Measurements of temporal coherence of sound transmissions through shallow water

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In this paper we report the measurements of temporal coherence of acoustic signals propagating through shallow water using data from three experiments in three different parts of the world, with sound speed standard deviation (STD) varying from 0.3 to 5 m/s near the layer depth. Temporal coherence is estimated from the autocorrelations of broadband channel impulse functions, the latter are deduced from broadband signals transmitted through the ocean during the experiments. The measurements covered three frequency bands: low frequencies below 1.2 kHz, midfrequencies between 2 and 5 kHz, and high frequencies between 18 and 22 kHz. The source-receiver range covers 3, 5, 10, and 42 km. The signal coherence-time is defined and deduced from the data. Motivated by previous theoretical work in deep water on the signal coherence-time as a function of the signal frequency, the source-receiver range, and sound speed STD, a similar but empirical analysis is applied to the measured data in shallow water. While the range dependence agrees with the theory, the data exhibit a different dependence on the signal frequency than the theoretical prediction for deep water. [DOI: 10.1121/1.2345910]

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## I. INTRODUCTION

The ocean is a dynamic (time varying/random) medium. As a result, acoustic signals propagating through the ocean fluctuate with time. The rate of the signal fluctuation is often described by the signal temporal coherence (i.e., the autocorrelation of the signal normalized by the average signal energy) that measures in a statistical sense the rate of decrease of the signal coherence. The higher the signal fluctuation, the faster the temporal coherence decreases with time. The signal coherence-time is defined as the time by which the signal becomes “uncorrelated” (as when the temporal coherence drops to  $1/e$ ).

Knowing the signal coherence-time is useful for many practical applications. For example, for the detection and localization of narrow band acoustic signals, the signal coherence-time determines the maximum time the signal can be integrated (before losing coherence), which, in turn, determines the maximum temporal processing gain. For matched field processing, the autocorrelation time between the initial field (used as the replica field) and the data field determines for how long the replica field can be used. For adaptive (temporal) signal processing, such as an adaptive channel equalizer in underwater acoustic communications, the rate of signal adaptation is controlled by the temporal stability of the signal or the signal coherence-time.

The practical need has prompted many experimental and theoretical studies of the temporal coherence of acoustic signals. Many experimental works reported in the literature were concerned with low frequency signals propagated to long ranges in deep oceans. Spindel *et al.*<sup>1</sup> and Clark *et al.*<sup>2</sup> focused on signal propagation at 406 Hz up to 250 km.

Williams<sup>3</sup> and Williams *et al.*<sup>4</sup> studied signal propagation at 400 Hz up to 1200 km. Georges *et al.*<sup>5</sup> and Birdsall *et al.*<sup>6</sup> investigated 57 Hz signal propagation to thousands of kilometers. Fitzgerald *et al.*<sup>7</sup> focused on very low frequency (10 Hz) signals. Dyer *et al.*<sup>8</sup> studied a band of frequencies (20–200 Hz) in the Marginal Ice Zone. The measurement results from these experiments can be expressed in terms of the signal phase rate, which is the inverse of the signal coherence-time, as discussed later. Temporal coherence was modeled theoretically using theories of signal propagations in a random medium. Earlier work calculated the signal phase rate using the Rytov approximation.<sup>9</sup> The emphasis was on the influence of internal waves in the deep ocean using the Garrett-Munk internal wave model.<sup>10</sup> The model predicted that the signal coherence-time is proportional to the inverse first power of frequency, inverse square-root power of range, and inversely proportional to the amplitude of the internal waves or the sound speed standard deviation (STD).

In contrast to the abundant work in deep water, there were few measurements on the temporal coherence of acoustic signals propagating in shallow water until recently. Using the ADVENT99 data collected in the Strait of Sicily, Nielsen *et al.* reported snapshot measurements of temporal coherence using linear frequency modulated (LFM) signals.<sup>11</sup> From the same data, using multitone signals, Yang *et al.* reported the temporal coherence of the acoustic signals using matched-field correlation of data fields.<sup>12,13</sup> By decomposing the data into mode space, they also measured the temporal coherence of individual (low) order modes. Both the LFM and multitone showed a signal coherence-time on the order of 20–200 min at frequencies below 1 kHz. Headrick *et al.* deduced the mode correlation from SWARM 95 data received on the WHOI vertical array at a range of 33 km.<sup>14</sup> Rouseff *et al.* measured and modeled the modal temporal coherence us-

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ing the data received on the NRL array at a range of 42 km.<sup>15</sup> Mignerey *et al.* reported a significant variation of the matched-field correlation of low frequency signals in the South China Sea, depending very much on the presence or absence of the solitary internal waves.<sup>16</sup> Badiey *et al.* measured the temporal coherence of acoustic signals in very shallow ( $\sim 15$  m) regions.<sup>17</sup>

Given the recent interest on signal propagation in shallow water, and the broadening of interest from low to high frequencies (such as for acoustic communications), the need exists for a systematic measurement of the temporal coherence of signals propagated through shallow waters to determine its relationship to the oceanographic environment. We report in this paper experimental measurements of temporal coherence in three different shallow water areas:<sup>18</sup> Sicily Strait in the Mediterranean, coastal water off New England, and coastal water off New Jersey, covering a frequency range from 200 Hz to 22 kHz and a propagation range from 3–4 km to 40 km. Issues critical to the signal coherence measurement are addressed.

From the measured signal coherence, one can estimate the signal coherence-time as defined later. A question of interest is what is the frequency and range dependence of the signal coherence-time in shallow water (as compared with deep water). For this study, an empirical analysis is used, noting the lack of theoretical predictions for shallow water. Another question of interest is what causes the signal coherence loss. In general, signal coherence loss arrives from sound scattering from the moving sea surface and time-varying oceanographic processes. (Signal coherence-time is infinite in a frozen ocean.) For a bottom mounted source in a downward refractive sound speed profile, acoustic propagation modeling results suggest that the contribution of the surface is minimal (see later). Assuming medium fluctuation is the dominant cause for the coherence loss, an effort is made to determine the relation between the signal coherence-time with the sound-speed STD, using conductivity-temperature-depth (CTD) data taken during the acoustic experiment.

To properly interpret the coherence-time results, one needs to model the influence of the various oceanographic processes (internal waves, turbulence, etc.) on the acoustic signal. Such a study is beyond the scope of this paper. Note that if the signal fluctuation can be described by one statistical distribution (as when there exists only one oceanographic process), the temporal coherence measurement will be independent of time and the size of the ensemble as long as the latter is statistically significant. In the real ocean, there exist various oceanographic processes that evolve with different time scales.

Consequently, the signal coherence measured at one time could be different than that measured at another time. Thus, in addition to the average temporal coherence of the signal, one needs to also study the “instantaneous” temporal coherence of the signal—the former is the average over samples covering a long period of time (hours/days), and the latter is the average over samples covering a short period of time (seconds/minutes). The “instantaneous” signal coherence-time provides a measure (variance) of the tempo-

ral variation of the signal coherence-time; the variance is a function of the acoustic frequency. The mean and variance of the “instantaneous” signal coherence-time can be used to infer the properties of the oceanographic processes responsible for the signal coherence loss.

The data analyzed involve acoustic signals of different carrier (center) frequencies with different bandwidths. Naturally, to compare the results, one needs to ask whether the measurements are influenced by the signal bandwidth. Note that the signal coherence for a (extra) wide band signal is not likely to be the same as that of a narrow band signal. When the signal bandwidth is small compared with the signal frequency, most works have assumed that the signal bandwidth is not an issue with which to be concerned. The bandwidth dependence is studied below by estimating the temporal coherence with a bandpass filter (with different bandwidths) applied to data with a 2 kHz bandwidth. The results showed little bandwidth dependence.

This paper is organized as follows. In Sec. II we describe the definitions and measurement techniques. In Sec. III we illustrate the measurements of the signal temporal coherence at three frequencies and two ranges using the ASCOT01 data, including a discussion on the measurement issues. In Sec. IV we present the results of signal temporal coherence measured from the SWARM95 experiment. In Sec. V we present the results of signal temporal coherence measured from RAGS03 experiment. The range and frequency dependence of the signal coherence-time are studied in Sec. VI. Environmental impacts on the signal temporal coherence are briefly discussed. Section VII is a short summary.

## II. DEFINITIONS AND MEASUREMENT TECHNIQUES

Temporal coherence of a signal is defined by the (auto-) correlation of the signals separated by a delay time  $\tau$ , normalized by the power of the signal, as given by

$$\rho(\tau) \equiv \frac{\langle p^*(t)p(t+\tau) \rangle}{\sqrt{\langle p^*(t)p(t) \rangle \langle p^*(t+\tau)p(t+\tau) \rangle}}, \quad (1)$$

where  $\langle \rangle$  denotes the ensemble average and  $*$  denotes the complex conjugate. The ensemble average is usually done over time  $t$  using the ergodic theorem. The ensemble average can also be over frequencies (i.e., different narrow band signals), or space (i.e., sensors over an array), or combinations of all the above.

Earlier work on deep water data was focused on the phase rate of a narrow band signal.<sup>1–8</sup> A narrow band signal can be represented after quadrature demodulation (i.e., after removing the carrier frequency) as

$$p(t) = R(t)e^{i\phi(t)}, \quad (2)$$

where  $R(t)$  is the amplitude and  $\phi(t)$  is the phase. It was assumed that the signal variation is predominantly in its phase. Ignoring the amplitude variation, i.e.,  $R(t+\tau) \approx R(t)$ , one finds that the autocorrelation of the pressure field can be approximated by<sup>19,20</sup>

$$\langle p^*(t)p(t+\tau) \rangle \simeq \mu^2 \langle e^{-i\phi(t)} e^{i\phi(t+\tau)} \rangle \simeq \mu^2 e^{(-1/2)\dot{\phi}^2 \tau^2}, \quad (3)$$

where  $\tau$  is the delay time,  $\dot{\phi}$  denotes the time derivative of the signal phase (phase rate), and  $\mu^2$  denotes the mean intensity  $\langle R(t)R(t+\tau) \rangle \simeq \mu^2$ . Using Eq. (3), the temporal coherence as defined by Eq. (1) becomes

$$\rho(\tau) = e^{(-1/2)\dot{\phi}^2 \tau^2}. \quad (4)$$

While noting that the assumptions made above for deep water may not be valid for shallow water, one observes at the same time that the functional dependence of the signal coherence [Eq. (4)] has been widely assumed for data analysis. It will be used later and seems to be valid for a small delay time.

The signal coherence-time, commonly defined by the delay time at which  $\rho(\tau) = 1/e$ , is given by

$$\tau_{1/e} = \sqrt{2}(\dot{\phi})^{-1}. \quad (5)$$

In terms of  $\tau_{1/e}$ , Eq. (4) becomes  $\rho(\tau) = e^{-\tau^2/\tau_{1/e}^2}$ . Defining  $\dot{\phi} = 2\pi\nu$ , one finds that  $\nu$  is the Doppler spread for a narrow band signal. This can be verified by taking a Fourier transform of the correlation function, Eq. (3). It is evident from Eq. (5) that the Doppler spread is inversely proportional to the signal coherence-time.

For reasons having to do with data analysis (as discussed later in Sec. III), one finds it necessary to measure the signal coherence-time while the signal coherence is high. A practical choice is to define  $\tau_{0.8}$ , which is the coherence-time at  $\rho = 0.8$ . It is related to the Doppler spread by

$$\nu = \sqrt{\frac{-\ln(0.8)}{2\pi^2}} \tau_{0.8}^{-1} \simeq 0.11 \tau_{0.8}^{-1}. \quad (6)$$

$\tau_{0.8}$  has been previously discussed in the context of underwater acoustic communications<sup>18</sup> as communications often require a signaling period with high signal coherence.

For deep water, using the Rytov approximation, earlier theoretical models have suggested that  $\nu$  should increase linearly with the acoustic frequency and square-root power of the source-receiver range,<sup>9</sup>

$$\nu = \gamma f \sqrt{R}, \quad (7)$$

where the coefficient  $\gamma$  is proportional to the amplitude of the internal waves. This suggests that the coherence-time  $\tau_{0.8}$  should decrease linearly with the inverse of the acoustic frequency and inverse square root of the range. This dependence will be tested with the shallow water data analyzed later.

A second method to measure the temporal coherence of a (narrow band) signal uses matched field correlation.<sup>12,16</sup> A (narrow band) signal field at time  $t$  is used as the reference (replica) field and correlated with the signal that follows at a later time,  $t + \tau$ . It is called matched field correlation since the ensemble average is over the sensors on an array,

$$\rho(t, \tau) = \frac{\sum_{j=1}^N p_j^*(t) p_j(t + \tau)}{\sqrt{\sum_{j=1}^N p_j^*(t) p_j(t) \sum_{j=1}^N p_j^*(t + \tau) p_j(t + \tau)}}, \quad (8)$$

where  $p(t)$  is the reference field. To achieve robustness, it is averaged over blocks of data at different time  $t$  or over signals of different frequencies, or both.

A third method to measure the signal temporal coherence uses broadband signals separated by a delay time  $\tau$ ,

$$\rho(t, \tau) = \left\langle \frac{[p^*(t) \otimes p(t + \tau)]_{\max}}{\sqrt{[p^*(t) \otimes p(t)]_{\max} [p^*(t + \tau) \otimes p(t + \tau)]_{\max}}} \right\rangle, \quad (9)$$

where  $[p^*(t) \otimes p(t + \tau)]_{\max}$  means the maximum value of the cross-correlation of the two time series or the convolution of the time-reversed signal (denoted by  $*$ ) with the other signal. A broadband signal is chosen as the reference signal and correlated with the signals arriving at a later time. The ensemble average will be over signals transmitted at different geotime  $t$ , i.e., different reference signals. Broadband temporal correlation can be shown to be equivalent to narrow band correlations summed over frequencies. Given enough data samples, these two measures should yield the same answer, at least in theory. This (third) method will be used extensively for data analysis in this paper.

As shown by Eq. (9), the signal temporal coherence is implicitly a function of  $t$ , the signal transmission time (or geotime). The measurement (averaged over data) near time  $t$  will be the “instantaneous” signal coherence-time. The measurement of the signal coherence-time could have a large variance depending on the signal frequency and the time scale of the oceanographic processes affecting the signal. The interpretation of the coherence-time data needs to be put in the context of the corresponding propagation environment; whatever models are used needs to reproduce the measured channel impulse responses first.

### III. ASCOT01 DATA ANALYSIS: THE MEASUREMENT ISSUES

In this section we analyze the data from the ASCOT01 experiments. It also addresses some of the measurement issues. For example, what are the suitable waveforms for temporal coherence measurement? What is the signal transmission rate to properly sample the temporal coherence? What is the effect of noise on the temporal coherence measurement? What is the minimum size of the samples for temporal coherence measurements? What is the effect of array motion on the measurements? Finally, how is the measurement influenced by the number of multipaths and the signal bandwidth?

The ASCOT01 experiment took place in June of 2001 off the coast of New England along the contour of 100 m depth.<sup>21,22</sup> It was part of joint SACLANTCEN and US experiments. The sound speed is nearly a constant with a velocity of 1465 m/s for depths below 30 m. The sound speed

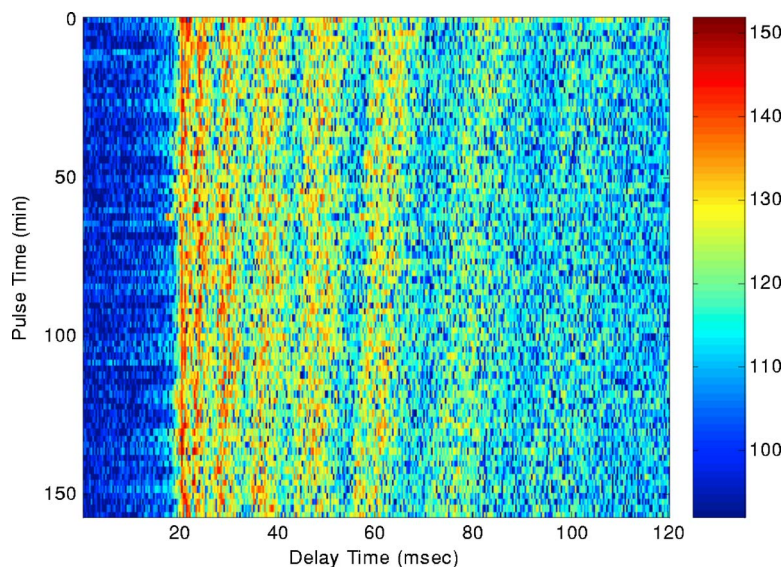


FIG. 1. (Color online) Measured channel impulse responses as a function of the delay time and geotime (time relative to the first pulse) based on LFM signals transmitted in the ASCOT01 experiment. Color scale is dB. Signal center frequency is 3550 Hz and source-receiver range is 10 km.

increases gradually from 30 m upward to 1500 m/s at 8 m depth. The source was mounted 4 m from the ocean bottom. A vertical line array was deployed at three distances of approximately 2, 5, and 10 km from the source. The data at 5 and 10 km will be used to study the range dependence of the temporal coherence of the signal. At each range, broadband phase-modulated signals were transmitted at three carrier frequencies, 450, 1200, and 3550 Hz. The data at different frequencies will be analyzed to study the frequency dependence of the temporal coherence. The signals at the lower two carrier frequencies had a bandwidth of 400 Hz. At 3550 Hz, signals were transmitted using several different bandwidths extending from 500 to 2000 Hz.

In the following subsections, the analysis focuses on data collected at the range of 10 km using three different carrier frequencies, followed by the data at 5 km. The receiver depth is between 65 and 82 m depth. No significant depth dependence is found in the temporal coherence result.

The signals analyzed were acoustic communication signals, transmitted in packets. Each packet lasted about 10 or 20 s depending on the signal frequency. A packet started with a LFM signal of 0.1 s, followed by a 0.3 s gap, followed by a Barker code signal of 13 symbols, followed by a 0.3 s gap, and then followed by the data signal. For the temporal coherence study, the data were BPSK signals coded with repetitive  $m$ -sequences. The  $m$ -sequences are “orthogonal” signals such that the signal autocorrelation is sharply peaked with a peak-to-sidelobe ratio of  $10 \log N$ , where  $N$  is the number of symbols used in the sequence. Unless otherwise mentioned, the signal packets were repeated every 2 min over a period of several hours. The LFM signal has the same bandwidth as the  $m$ -sequence signal.

LFM signals have been commonly used in the past to measure the channel impulse response functions and estimate the signal temporal coherence. To allow the reverberation to die down, the pulses were usually transmitted on the order of minutes to tens of minutes. It should become clear below that for channels with a short coherence-time, the LFM signal is a poor choice as the channel is too infrequently sampled. This is particularly true at high frequencies. To sample the

channel at a high sampling rate,  $m$ -sequence signals are used, transmitted in sequence continuously for an extended period of time. The  $m$ -sequence signals sample the channel impulse responses at a period of  $N/\text{bandwidth}$  s.

#### A. 3550 Hz at 10 km range

The channel impulse responses were obtained by matched filtering the data with the transmitted signal. The channel impulse responses from the LFM signals are plotted in Fig. 1 displaying the multipath arrivals as a function of delay time and geotime (transmission time); the LFM signals were transmitted every 2 min. The signal bandwidth is 2 kHz. One clearly sees the variation of the channel impulses between packets. One notes in Fig. 1 eight groups of multipath arrivals, with each group consisting of many micropath arrivals. The micropath arrivals are rays that scatter from the ocean inhomogeneities producing microrays. As a result, the arrival times of the micropath arrivals are spread out and their intensities fluctuate with time. The experimental site (in the New England coastal water) is known to have internal waves and turbulence.<sup>22</sup> The effect of these oceanographic processes on the acoustic signal propagation is evident in Fig. 1, particularly the late arrivals.

(One notes a gradual decrease in the delay time in Fig. 1 between the early and late arrivals as geotime increases. The shift in the arrival time of the late arrivals can be explained in terms of the internal tides, which shift the depth of the channel thermocline.<sup>23</sup> See Sec. VI D for further details).

Take one impulse response function as a reference and cross correlate with the impulse responses transmitted at a later time. The maximum values of the correlations are entered in Eq. (9) and one obtains one temporal coherence curve, as shown in Fig. 2(a). Take different impulse response functions as the references; one obtains a distribution of temporal coherence curves at different geotimes, as shown in Fig. 2(a). The average of all the curves is shown in Fig. 2(b), displayed by the data points marked as crosses. One observes that the average coherence decreases to a value of  $\sim 0.25$  over a period of 5–20 min.

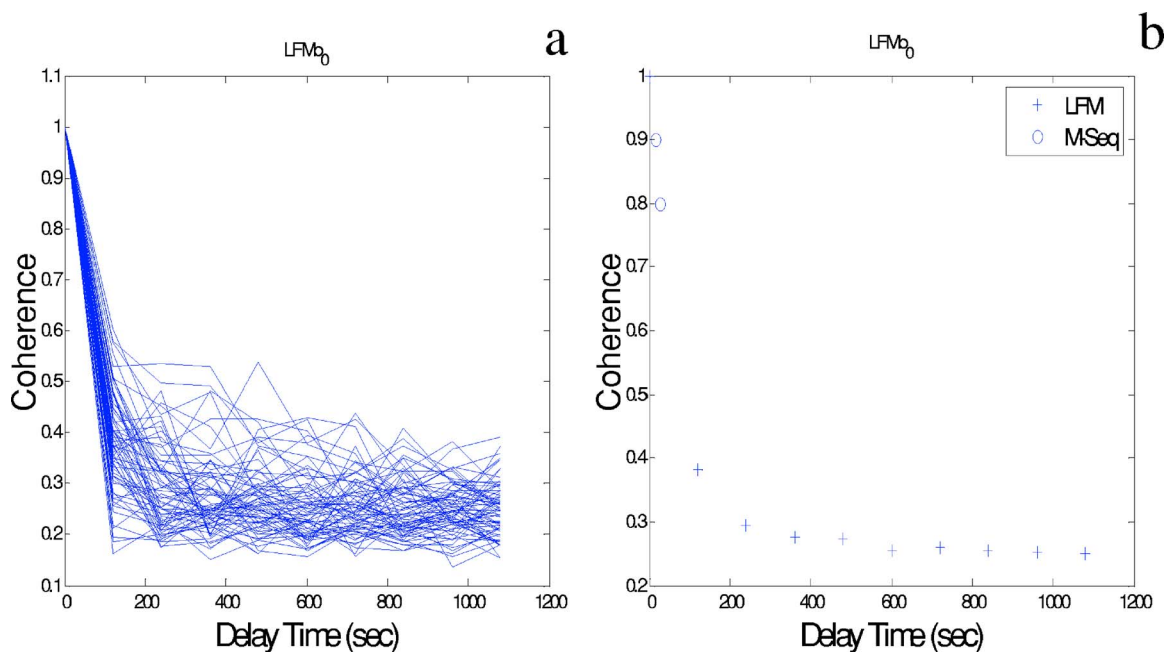


FIG. 2. (Color online) Temporal coherence curves measured at a different starting time (a), and averaged over 2.5 h of data using LFM signals (b). This figure shows that the normal method using LFM pulses is inadequate for measuring the temporal coherence at a short delay time. Superimposed in Fig. 4(b) is the measurement from *m*-sequence data from Fig. 4. Center frequency = 3550 Hz, and range = 10 km.

Figure 2(a) shows a wide spread in time for a fixed coherence value. For example, for a coherence value at  $1/e$ , the coherence-time can vary from 120 s to greater than 1000 s. Thus, the conventional definition of coherence-time, defined by the time delay at which the coherence value drops to  $1/e$ , could not be applied directly to this data. In general, the variance of the coherence-time at a fixed coherence value increases significantly as the coherence value decreases. In this paper, the measurement of the signal coherence-time is based on the time when the coherence drops to 0.8, i.e.,  $\tau_{0.8}$ . The reason is to keep the STD of the (coherence-time) measurements smaller than the average (coherence-time). The coherence-time  $\tau_{1/e}$  can be extrapolated from  $\tau_{0.8}$  given a model of temporal coherence.

Figure 2(b) shows that the coherence value drops to be-

low 0.5 at the first sample, the 120 s delay time. To measure  $\tau_{0.8}$ , it is necessary to increase the sampling rate using a shorter sampling interval. To this end, the *m*-sequence data were analyzed next. Each *m* sequence contained  $N=1023$  symbols and lasted 0.51 s. The *m* sequences were repeated for a total duration of 17 s.

The *m*-sequence data were processed the same way as the LFM signals. Figure 3 shows the channel impulse response for *m* sequences for one packet of data. Because the *m*-sequence signals are nearly orthogonal, with a sidelobe level that is, in principle,  $10 \log N$  dB down from the main level, the (circular) matched-filter correlation of the *m*-sequence data yields a good measurement of the channel impulse response function. Note that the amplitudes and arrival time are pretty stable for multipath arrival with a delay

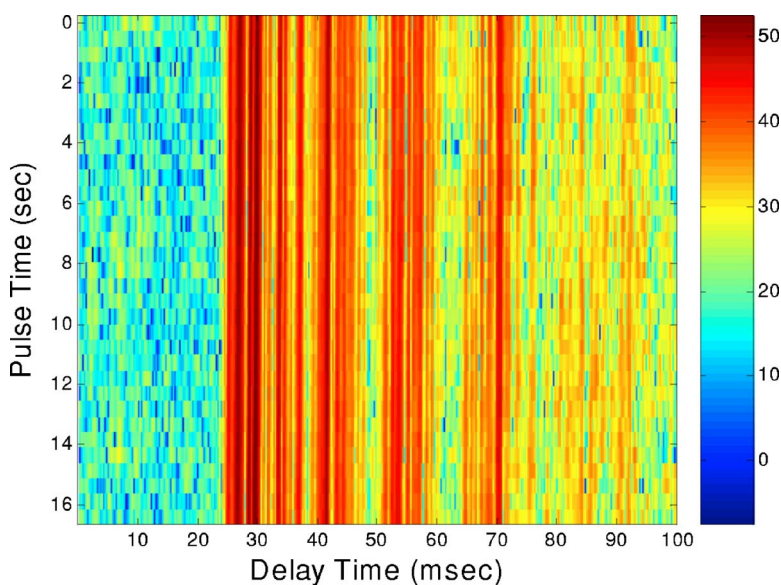


FIG. 3. (Color online) A typical display of the channel impulse responses measured from consecutive *m* sequences within a packet as a function of the delay time and geotime. Center frequency is 3550 Hz and range is 10 km. Color scale is dB. The *m* sequences are commonly used to characterize the channel variation within a packet of acoustic communication signal.

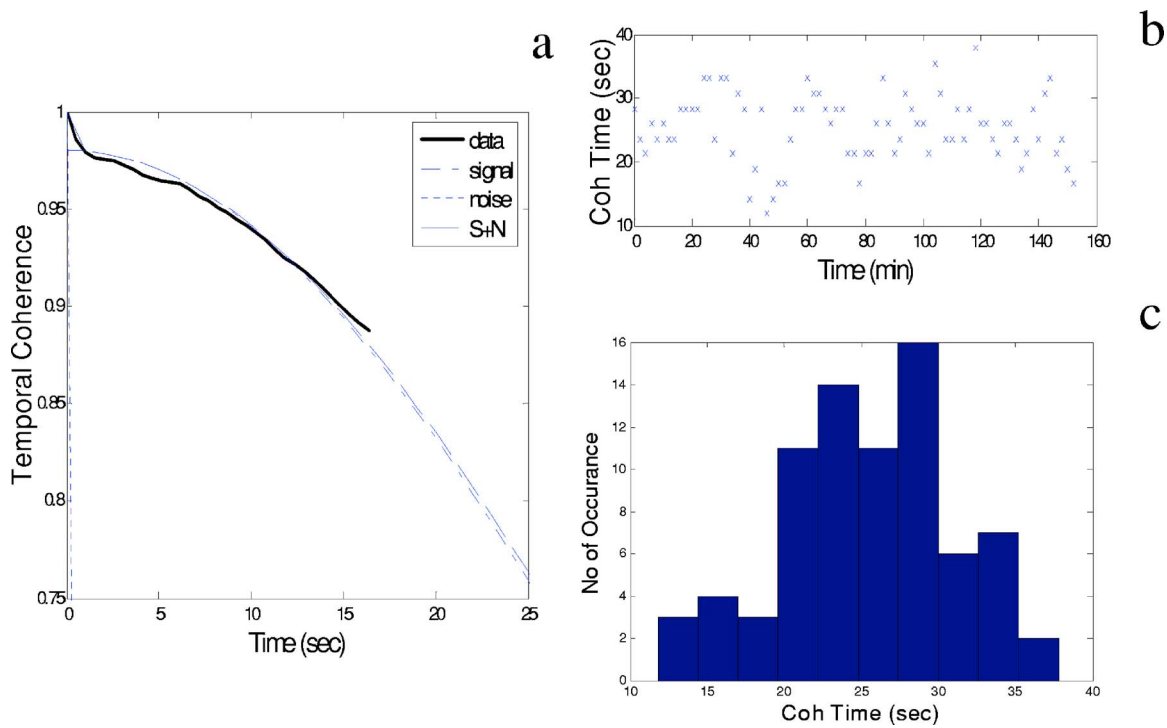


FIG. 4. (Color online) Average temporal coherence of the  $m$ -sequence signals, compared with a model of signal and noise (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c). Center frequency=3550 Hz, range=10 km.

time of less than 50 ms. Significant amplitude fluctuations are noted for arrivals with a delay time greater than 50 ms. This implies that the  $m$ -sequence signals are highly coherent within a packet.

Figure 4(a) shows the temporal coherence of the  $m$ -sequences averaged over 77 packets of data (the solid curve), covering a time span of 2.5 h. Note a sharp decrease of the measured temporal coherence near the origin with a slope very different from the rest of the curve. To explain this, one needs to include the contribution of noise in Eq. (1). Denoting the signal as  $s$  and noise as  $n$ , thus  $p=s+n$ . The measured temporal coherence  $\rho(\tau)$  is found to be

$$\rho(\tau) = s(\tau) \frac{S^2}{S^2 + N^2} + n(\tau) \frac{N^2}{S^2 + N^2}, \quad (10)$$

where  $s(\tau)$  is the temporal coherence of the signal and  $n(\tau)$  is the temporal coherence for the noise. In deriving Eq. (10), it is assumed that the signal and noise are uncorrelated. The coefficients multiplying the signal and noise coherence terms in Eq. (10) are the signal to signal-plus-noise ratio and noise to signal-plus-noise ratio.

One notes that noise is normally assumed uncorrelated in the temporal space. It has in reality a very short coherence-time. The sharp drop of the measured coherence in Fig. 4(a) at a short delay time can be attributed to the noise, as illustrated later. The signal is expected to have a coherence-time much longer than the noise. Consequently, the signal coherence curve should have a slower (absolute) slope than the noise coherence curve. The data with a delay time greater than 1 s are that associated with the signal. One can fit the data in Fig. 4(a) using Eq. (10), assuming that both the signal and noise coherence are modeled by Eq. (6). The

modeled signal and noise coherence curves are shown in Fig. 4(a) as the dash-dotted and dotted lines, respectively, assuming  $\tau_{le}=50$  and 0.1 s for the signal and noise, respectively. (The signal coherence curve is displaced by 0.02 from unity.) The sum of the signal plus noise coherence using Eq. (10) is shown as the dashed line and is found in good agreement with the measured coherence data (solid curve). In fitting the data, one assumes a value of 0.02 for the noise to signal-plus-noise ratio, which is estimated by the contribution of the second term in Eq. (10) at zero delay time. This yields a signal-to-noise ratio of 17 dB, which agrees with the estimation based on the raw data.

The data analysis procedure is to determine the signal coherence from the measured coherence data by removing the noise contribution, which concentrates at a delay time less than 1 s. From the signal temporal coherence, one then estimates the signal coherence-time. From the data shown in Fig. 4(a), one finds the coherence-time of the average signal coherence at a coherence value of 0.8 is about 23 s. One can also obtain the “instantaneous” coherence-time of the signal by applying the above procedure to the temporal coherence curve measured from each packet of the  $m$ -sequence data. The measured “instantaneous” coherence-time of the signal is plotted in Fig. 4(b) as a function of geotime of the transmitted packet. Figure 4(c) shows the histogram distribution of the “instantaneous” coherence-time of the signal. From the histogram one obtains a mean coherence-time  $\tau_{0.8}$  of around 25 s, approximately the same as the coherence-time of the average signal coherence reported above. One also finds a STD of  $\sim 5$  s for the mean coherence-time measurement.

The above data were also used to study the bandwidth

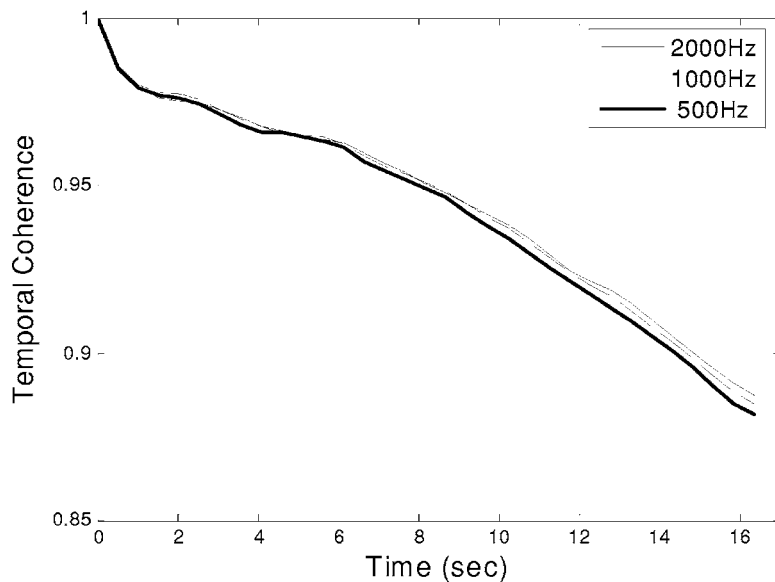


FIG. 5. Bandwidth dependence of the average temporal coherence of the signal. Center frequency=3550 Hz, range=10 km.

dependence of the signal temporal coherence. In general, the signal coherence will depend on the signal bandwidth; a wideband signal will have a different temporal coherence than a narrow band signal. In addition to sensing different frequency responses of the propagation channel, a wide band signal can resolve individual (group of) path arrivals and measure the temporal coherence of these paths, which a narrowband signal cannot. To conduct the study, signals of different bandwidths are obtained by applying a bandpass filter to the raw data (mentioned above) using a different filter bandwidth. Temporal coherence is next deduced from the filtered data. Figure 5 shows the measurements of the temporal coherence of the bandpassed signals. One notes very little dependence of the temporal coherence on the signal bandwidth. This result is perhaps expected when the bandwidth of the signal is small compared with the signal carrier frequency, as is the case for the 3550 Hz data. The lack of bandwidth dependence as demonstrated by the 3550 Hz signal will be assumed to be true for signals at other frequencies. This allows us to compare and interpret data of different

bandwidths as presented later without considering the bandwidth as a separate parameter in the equation.

### B. 1200 Hz at 10 km range

The channel impulse responses measured from the LFM signals are plotted in Fig. 6. The signals were transmitted every 2 min covering a period of 8 h. The LFM signals had a bandwidth of 400 Hz. Consequently, the temporal resolution in Fig. 6 is not as good (sharp) as that seen in Fig. 1. Nonetheless, one still sees eight groups of multipath arrivals. Figure 6 shows clearly that the multipath arrival time (particularly the last four groups of multipaths) also varies with geotime. The gradual shift of the late arrivals with geotime is similar to that shown in Fig. 1 and can be explained by the shift of the thermocline axis due to (semidiurnal) internal tides.

The impulse responses at a different geotime were used as the references and cross-correlated with the impulses responses that follow. For each reference, a temporal coher-

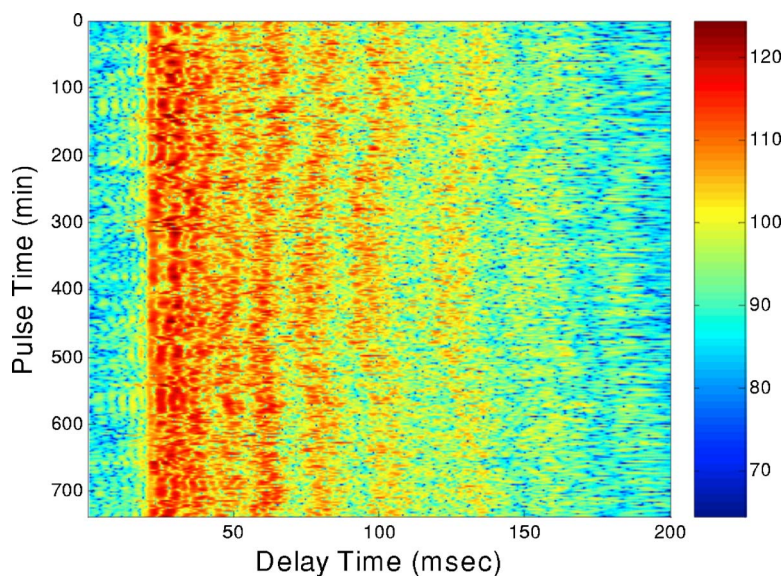


FIG. 6. (Color online) Channel impulse responses at 1200 Hz as a function of the delay time and geotime, based on LFM data received at a range of 10 km during the ASCOT01 experiment. Color scale is dB. The multipath arrival pattern is similar to that shown in Fig. 1.

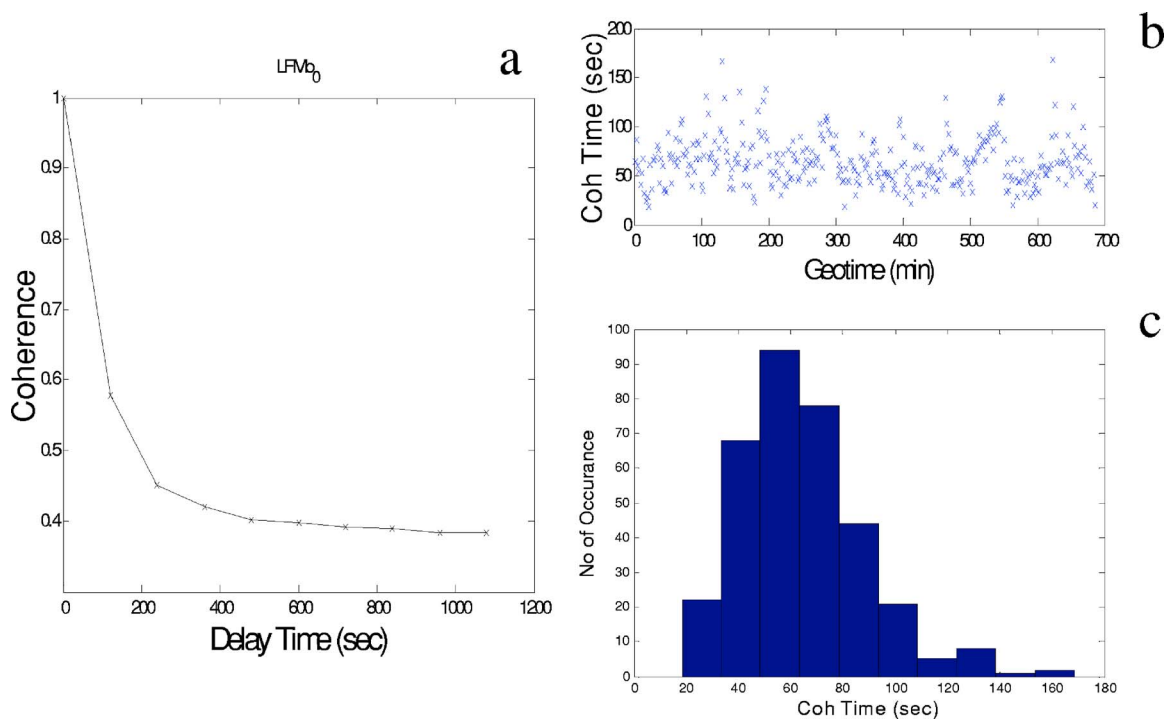


FIG. 7. (Color online) Temporal coherence measured from data shown in Fig. 6. Average temporal coherence as a function of signal separation time (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c).

ence curve was obtained. Figure 7(a) shows the average of all the temporal coherence curves. From this, one can estimate the coherence-time of the average signal coherence to be  $\tau_{0.8} \approx 57$  s. One can also estimate the coherence-time  $\tau_{0.8}$  for each of the “instantaneous” temporal coherence curves obtained. The results are plotted in Fig. 7(b) to show the temporal variation (time dependence) of the “instantaneous” signal coherence-time. It indicates a cycling period of 30–50 min, which is shorter than the periods of the internal tides. The cause of the temporal variation of the signal coherence-time has not been investigated.

Figure 7(c) shows the histogram of the measured coherence-time. One finds a mean of approximately 64 s and a STD of about 24 s.

### C. 450 Hz at 10 km range

The 450 Hz signals had a bandwidth of 400 Hz covering a frequency range of 250 to 650 Hz. Figure 8 shows the measured channel impulse responses deduced from the LFM signals. The signals, transmitted every 2 min, covered a period of 3.5 h. The impulse responses at 450 Hz are clearly different from the impulse responses shown earlier at 1200 or 3550 Hz. Although seven to eight groups of multipath arrivals can also be identified in Fig. 8, one finds that the second and fifth multipaths are the dominant arrivals versus the first to third arrivals in Figs. 1 and 6.

The impulse responses were processed to produce the temporal coherence curves at a different geotime (as de-

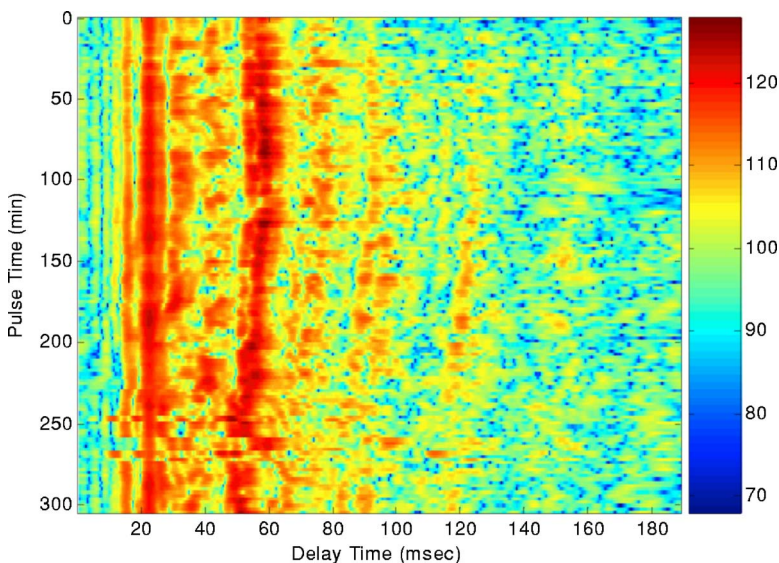


FIG. 8. (Color online) Channel impulse responses at 450 Hz as a function of the delay time and geotime based on LFM signals transmitted during the ASCOT01 experiment. Color scale is dB. Source-receiver range is 10 km. Second and fifth arrivals have higher amplitudes than the others. Impulse responses are aligned by the peak amplitude within the first 30 ms of data; data at geotime from 240 to 270 min are misaligned by this method.

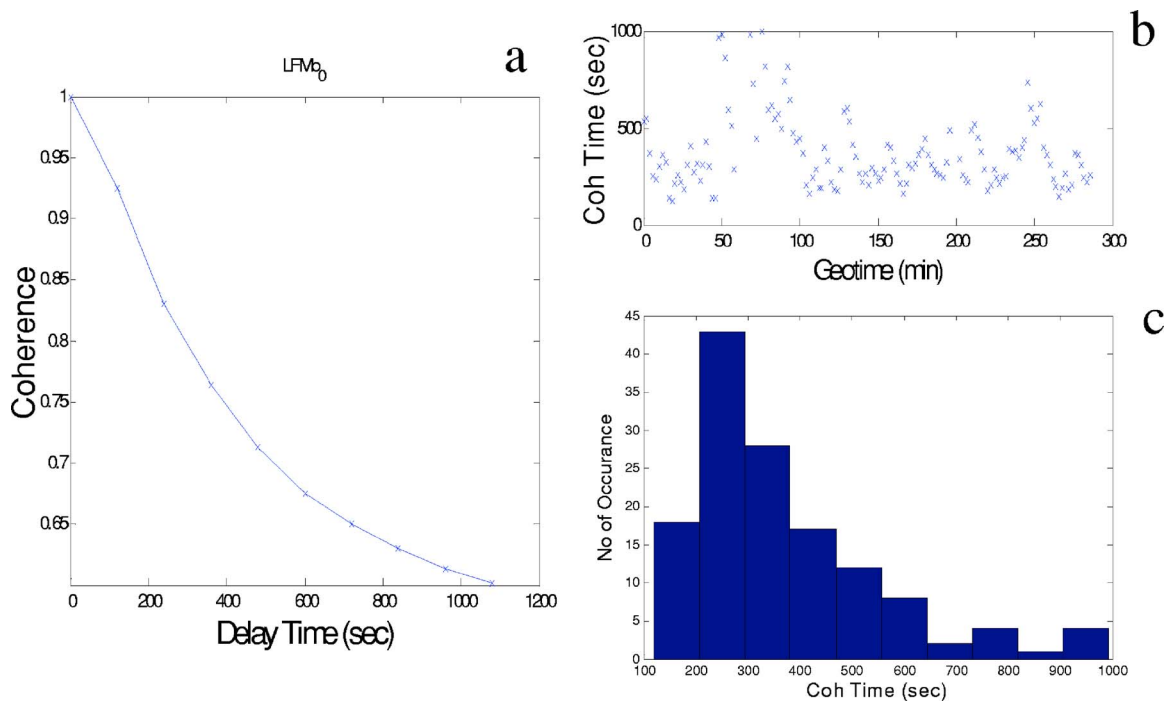


FIG. 9. (Color online) Temporal coherence based on data of Fig. 8. Average temporal coherence as a function of signal separation time (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c).

scribed above). Figure 9(a) shows the average of all the temporal coherence curves. One finds the coherence-time  $\tau_{0.8}$  of the average signal coherence to be about 299 s. The coherence-time estimated from each of the temporal coherence curves (described above) is plotted in Fig. 9(b). Figure 9(b) shows the variation (time dependence) of the “instantaneous” signal coherence-time. Figure 9(c) shows the histogram of the measured coherence-time. One finds a mean  $\tau_{0.8}$  of approximately 382 s and a STD of about 202 s. The high value of the STD arises from the high “instantaneous” coherence-time during the period of 50–90 min in Fig. 9(c), which is distinctly different from the rest of the time. There is no apparent correlation between the time series of the “instantaneous” coherence-time [Fig. 9(b)] and the multipath arrival time (Fig. 8); the later is related to the tidal cycle.

Note that the STD measured over a short period of time (10–40 min) is much smaller ( $\sim 100$  s). For a smaller time period (minutes), the STD will be smaller. Thus the STD in the coherence-time could be used as an indicator of the amount of the changes in the acoustic environment (e.g., the sound speed STD) during the measurement periods. See Sec. VI for further study.

#### D. Data at 5 km range

The data at the 5 km range were processed in the same manner as the data at the 10 km range. Only LFM signals were used. The results for 3550, 1200, and 450 Hz are shown in Figs. 10–15, respectively.

The measured channel impulse responses at 3550 Hz are

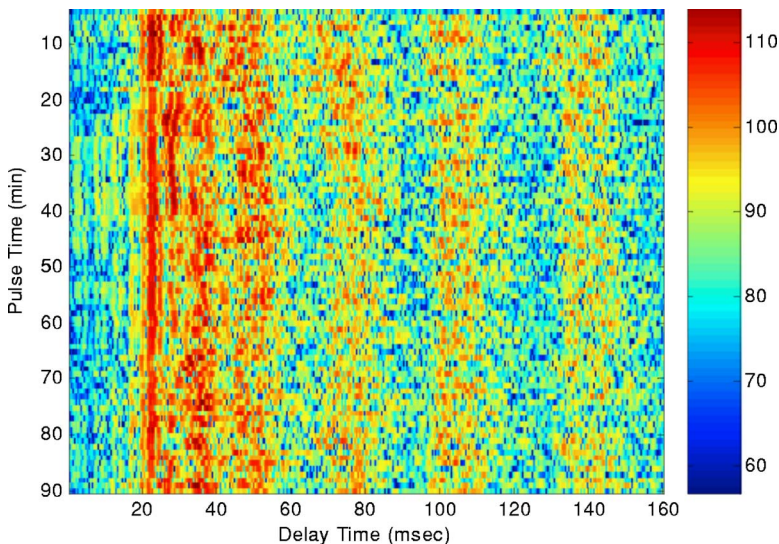


FIG. 10. (Color online) Channel impulse responses at a range of 5 km as a function of the delay time and geotime based on LFM signals centered at 3550 Hz from the ASCOT01 experiment. Color scale is dB. Similar arrival pattern as in Fig. 1 but with a nearly doubled multipath spread.

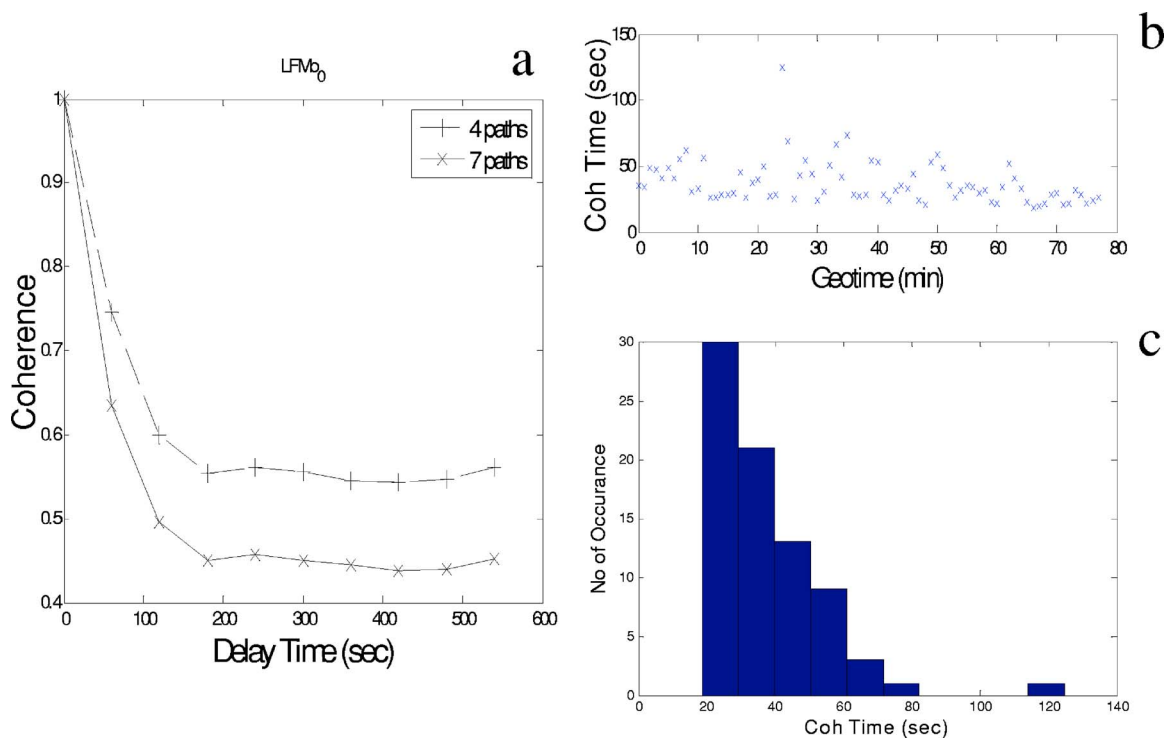


FIG. 11. (Color online) Temporal coherence measured from data of Fig. 10. Average temporal coherence using the impulse responses containing all seven paths versus only the initial four paths (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c). Center frequency=3550 Hz, range=5 km.

shown in Fig. 10 as a function of the delay time and geotime. The difference between this data (Fig. 10) and the data at 10 km (Fig. 1) is that the 3550 Hz signals at 5 km had only a bandwidth of 500 Hz. The signals, transmitted every 1 min, covered a period of 1.5 h. The channel impulse responses, deduced from the LFM signals at 1200 Hz, are plotted in Fig. 12. The signals, transmitted every 2.5 min, covered a period of 4 h. The channel impulse responses deduced from the LFM signals at 450 Hz are plotted in Fig. 14. The signals, transmitted every 2.5 min, covered a little more than 3 h. Both the 1200 and 450 Hz signals have a 400 Hz bandwidth, the same as the 10 km case.

One observes seven to eight groups of multipaths in

Figs. 10, 12, and 14. Certain groups of arrivals are clearly split into two subgroups of arrivals in Figs. 12 and 14. Comparing Figs. 10, 12, and 14 with Figs. 1, 6, and 8, one finds that the spreads of the multipath delay times (from the first arrival) at 5 km are almost double that at 10 km. The peak intensity distribution of the 450 Hz impulse responses changes significantly from 10 to 5 km.

The impulse responses were processed (as described above) to produce the temporal coherence curves at a different geotime—using the impulse response at a different time as the reference. coherence-time  $\tau_{0.8}$  was estimated from each of the temporal coherence curves.

Figure 11(a) shows the average temporal coherence of

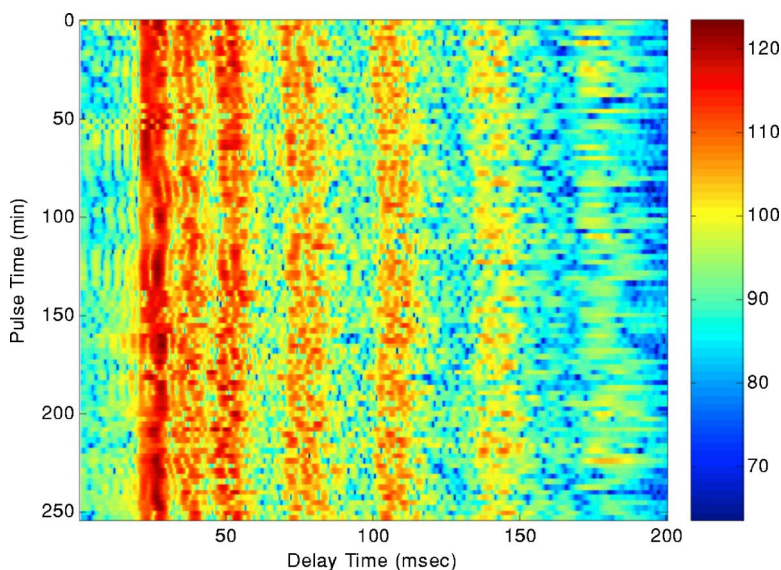


FIG. 12. (Color online) Channel impulse responses as a function of the delay time and geotime at a range of 5 km and center frequency of 1200 Hz based on LFM signals from the ASCOT01 experiment. Color scale is dB. Similar multipath arrival pattern as in Fig. 6, but with a nearly doubled delay spread.

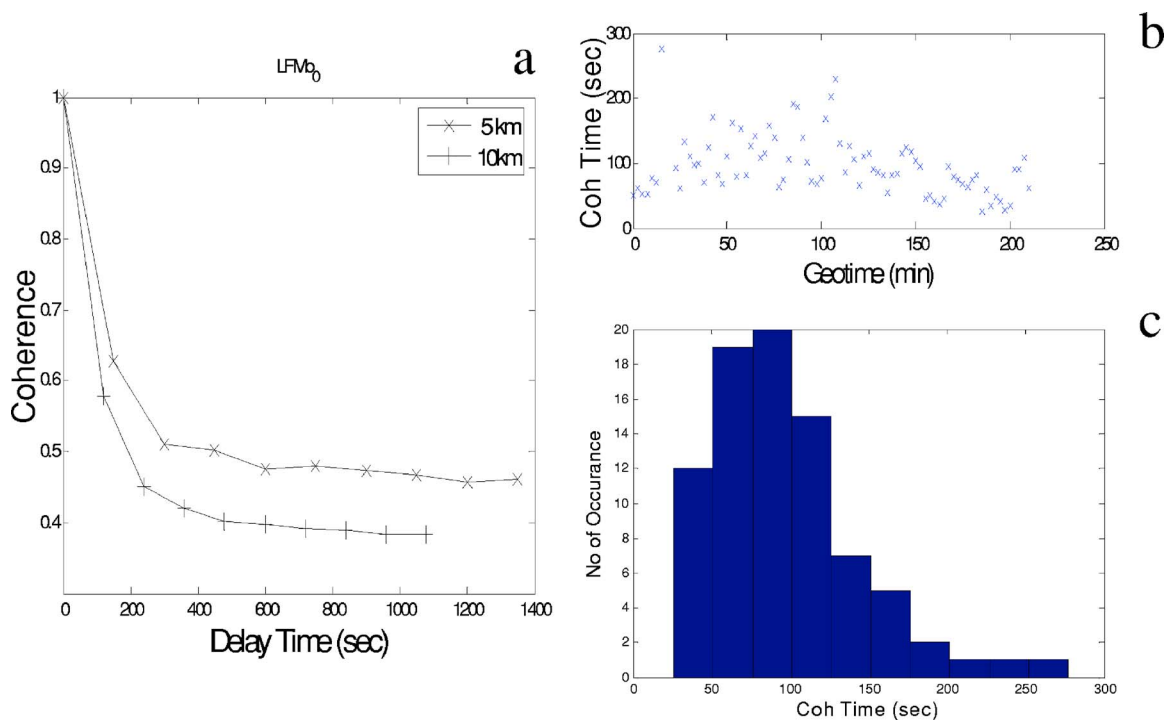


FIG. 13. (Color online) Temporal coherence measured from the data of Fig. 12. The average temporal coherence as a function of the signal separation time (a). "Instantaneous" signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c). The average temporal coherence of the signal at 10 km [Fig. 7(a)] is also displayed in Fig. 13(a) for comparison.

the signal at 3550 Hz (the solid curve). One finds the coherence-time  $\tau_{0.8}$  of the average signal coherence to be about 33 s. Figure 11(b) shows the temporal variation (time dependence) of the "instantaneous" signal coherence-time at 3550 Hz. Figure 11(c) shows the histogram of the measured coherence-time. One finds a mean of approximately 37 s and a STD of about 16 s for the 3550 Hz signal at 5 km.

The signal coherence-time of the 1200 Hz signal at 5 km is shown in Fig. 13. Figure 13(a) shows the average of all the temporal coherence curves (the solid curve) at 5 km. One finds the coherence-time  $\tau_{0.8}$  of the average signal coherence to be about 81 s. Superimposed is the previously obtained average temporal coherence curve at 10 km (the

dashed curve). Figure 13(b) show the temporal variation (time dependence) of the "instantaneous" signal coherence-time at 1200 Hz. Figure 13(c) shows the histogram of the measured coherence-time. One finds a mean of approximately 96 s and a STD of about 46 s.

The signal coherence-time of the 450 Hz signal at 5 km is shown in Fig. 15. Figure 15(a) shows the average of all the temporal coherence curves (the solid curve). One finds the coherence-time  $\tau_{0.8}$  of the average signal coherence to be about 443 s. Superimposed is the previously obtained average temporal coherence curve at 10 km (the dashed curve). Figure 15(b) shows the temporal variation (time dependence) of the "instantaneous" signal coherence-time at 450 Hz. Fig-

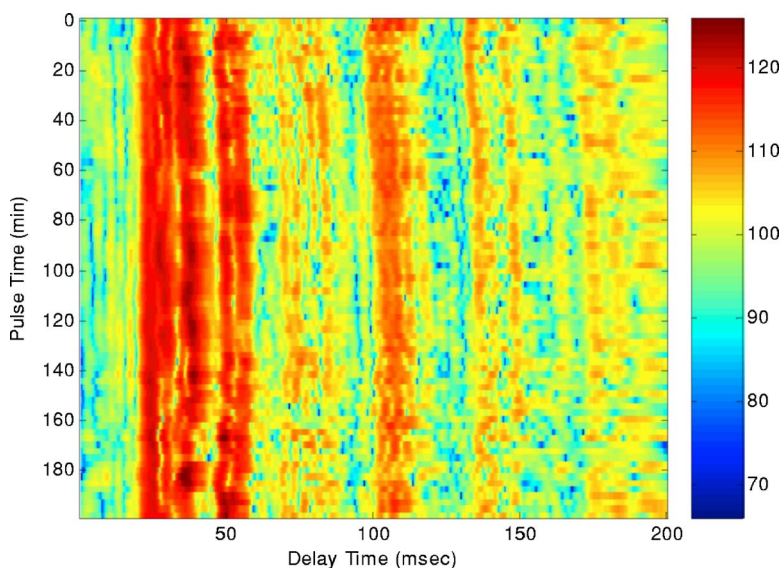


FIG. 14. (Color online) Channel impulse responses as a function of the delay time and geotime at a range of 5 km and center frequency of 450 Hz based on LFM signals from the ASCOT01 experiment. Color scale is dB. Multipath delay spread is almost twice that of Fig. 8, but the amplitude distribution is different.

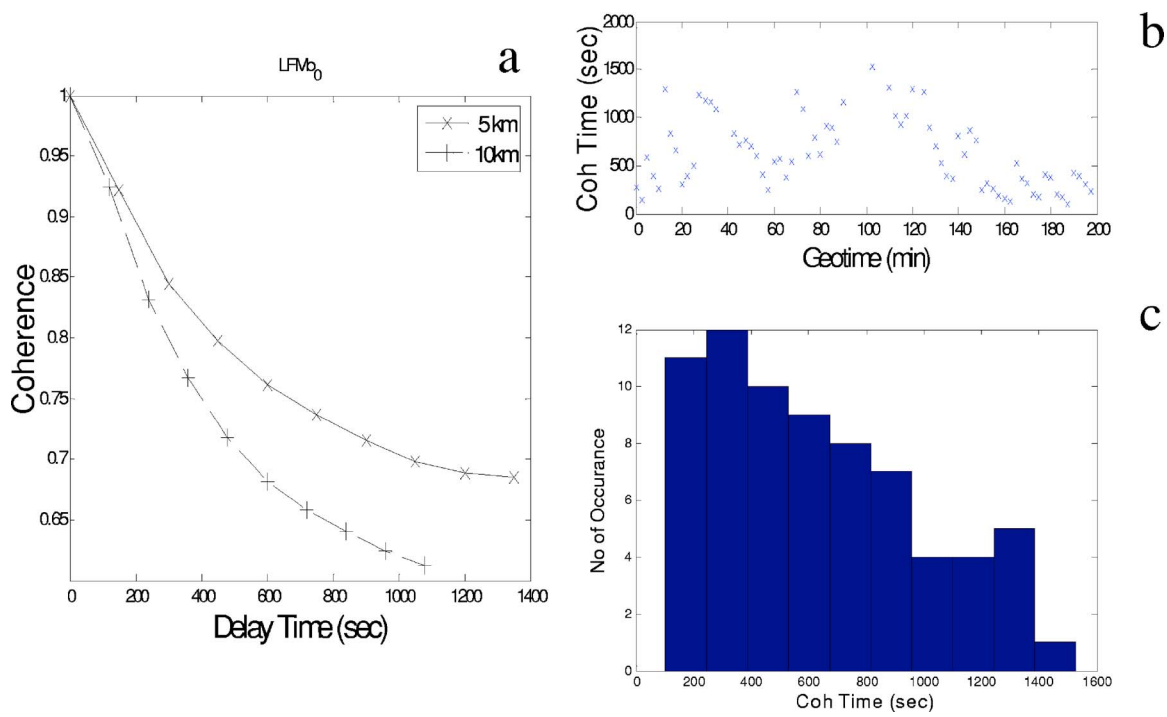


FIG. 15. (Color online) Temporal coherence measured from data of Fig. 14. Average temporal coherence of the LFM signals as a function of signal separation time (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c). The average temporal coherence of the signal at 10 km [Fig. 9(a)] is also displayed in Fig. 15(a) for comparison.

ure 15(c) shows the histogram of the measured coherence-time. One finds a mean of approximately 626 s and a STD of about 368 s.

### E. Comments on the coherence measurements

As is well known, the temporal coherence of the signal may vary from path to path. In general, the early arrivals are more coherent than the latter (scattering) arrivals. Hence, measurement of the temporal coherence could depend on the number of multipaths included in the signal. To illustrate this, it is plotted in Fig. 11(a) the average temporal coherence of the signal using only (the impulse responses of) the first four multipaths (the dashed curve) to compare with that using all seven multipaths (solid curve). One finds that the temporal coherence value increases by  $\sim 20\%$  at elapsed time  $> 100$  s.

The measurements of the signal temporal coherence in this paper include all the “dominant” multipath arrivals. To do so, one must use a sufficiently long time window for measuring the impulse response delay spread, yet not too long to increase the noise contribution. The procedure used here is to increase the window until the (average) coherence value converges to within a certain tolerance. The final time-window-used is the minimum time window-that achieves the convergence value.

One observes that the mean of the coherence-times is generally somewhat higher than the coherence-time estimated from the average of the temporal coherence curves. Note that the former is the average of the (“instantaneous”) coherence-time at a fixed coherence value, whereas the latter is deduced from the average of the coherence at fixed elapsed times. Note that the measurements are only available at dis-

crete elapsed times. Hence measurement data are interpolated or extrapolated to determine the coherence-time  $\tau_{0.8}$ . The details of interpolation and extrapolation, which involves curve fitting, could contribute to the difference between the two different ways of measuring the coherence-time, particularly when the “instantaneous” temporal coherence exhibit high degrees of fluctuations. It is observed in Fig. 2(a) that the variance of the data at a fixed coherence value is much higher than the variance of the data at a fixed elapsed time. Consequently, it is assumed that the coherence-time estimated from the average of the temporal coherence curves is more trustworthy than the mean of the coherence-times. The coherence-time estimated from the average of the temporal coherence curves is entered as the coherence-time  $\tau_{0.8}$  in Table I. The STD measured above, which represents the spread (or the temporal variation) of the “instantaneous” coherence-time, is also entered in Table I. This STD represents the environmental influence on the signal coherence-time. It should not be confused with the STD of the measurements (or the measurement error due, for example, to the different sizes of data samples). Note that the measurements of the temporal coherence for a broadband signal use an

TABLE I. coherence-time  $\tau_{0.8}$  and the spread of the coherence-time (in parentheses) measured from ASCOT01 data; the later reflects the amount of the temporal variation of the coherence-time due to changes in the propagation environment.

| Frequency | $\tau_{0.8}$ at 10 km | $\tau_{0.8}$ at 5 km |
|-----------|-----------------------|----------------------|
| 3550 Hz   | 23 ( $\pm 5$ ) s      | 33 ( $\pm 16$ )      |
| 1200 Hz   | 57 ( $\pm 24$ ) s     | 81 ( $\pm 46$ )      |
| 450 Hz    | 299 ( $\pm 202$ ) s   | 443 ( $\pm 368$ )    |

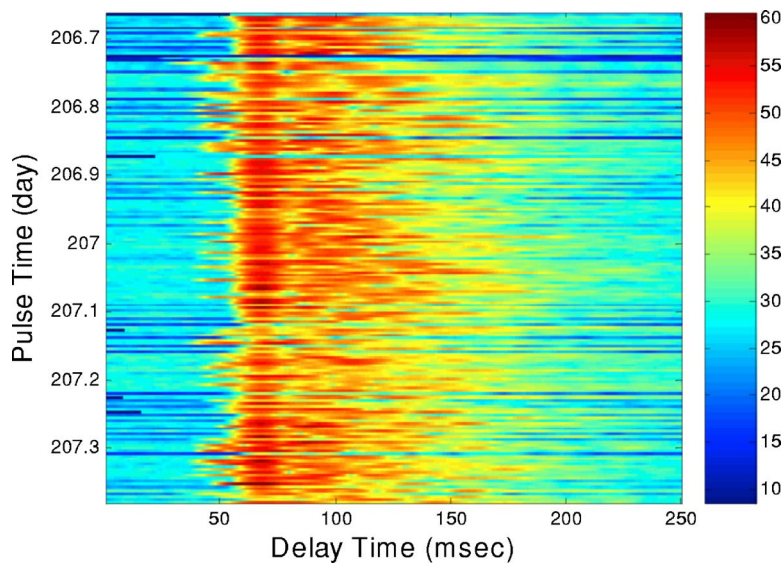


FIG. 16. (Color online) Results from the SWARM95 experiment. Measured channel impulse responses as a function of the delay time and geotime, showing groups of arrivals. Color scale is dB. Data based on *m*-sequence signals centered at 400 Hz with a range at 42 km.

average over frequencies. In contrast, the temporal coherence for a narrow band signal is averaged over data at different time. Both use an adequate number of samples.

#### IV. SWARM 95 DATA ANALYSIS

In this and the next section, the above analysis is applied to data from another site, off the coast of New Jersey. Although the measurement procedures are the same and repetitive as before, the data features (the delay structure and time variations of the channel impulse responses) are very different. How is signal coherence-time related to the channel dynamics is of great interest and will be briefly discussed in Sec. VI.

The SWARM 95 experiment was conducted in July–August of 1995 in the Mid-Atlantic bight continental shelf region east of New Jersey. It was sponsored by the Office of Naval Research with participation from many institutions. Details of the experiment were described in a previous publication.<sup>24</sup> The data to be analyzed here are data from two acoustic sources located upslope in 55 m of water to the NRL vertical line array down slope in 88 m of water. The source receiver range was  $\sim 42$  km. A 400 Hz source was deployed at midwater depth and a 224 Hz source was deployed near the bottom. The 400 Hz signals had a bandwidth of 100 Hz and the 224 Hz signals had a bandwidth of 16 Hz. The NRL vertical line array (VLA) contained 32 elements spanning the water column from 21 to 85 m.

BPSK data using *m* sequences were analyzed following the analysis described above. The BPSK signal at 400 Hz used an *m* sequence of 511 symbols repeated 23 times. Each packet lasted 117.53 s and was repeated continuously with a gap of 242.47 s in between packets. The BPSK signal at 224 Hz used an *m* sequence of 63 symbols repeated 30 times. Each packet lasted 118.13 s and was repeated continuously with a gap of 181.9 s in between packets.

A total of 179 packets was analyzed for the 400 Hz signal, yielding 4117 measurements of channel impulse responses covering a period of 17.9 h. The channel impulse responses were plotted in Fig. 16 for a receiver at 51 m depth. Temporal coherence was estimated for each packet

using the first *m* sequence as the reference. The average of all the coherence curves is plotted in Fig. 17(a). It yields a coherence-time  $\tau_{0.8} \approx 40$  s. The “instantaneous” coherence-time estimated from each packet is shown in Fig. 17(b). Figure 17(c) shows the histogram distribution of the “instantaneous” coherence-time. One finds a mean coherence-time of 55 s and STD of 40 s. The high STD is a manifestation of the (significant) environmental change in the SWARM95 area due to the presence of internal waves.

For the same period, the 224 Hz data yield 6450 measurements of channel impulse responses, which are plotted in Fig. 18(a). It showed one dominant group of arrivals. The mean temporal coherence averaged over all packets is plotted in Fig. 18(b), from which one finds the coherence-time  $\tau_{0.8} \approx 130$  s. In this case, one finds significantly larger fluctuations in the measurements of the “instantaneous” coherence-time than in the case of 400 Hz signal. For many packets, the coherence-time is beyond the packet length of 118 s and cannot be reliably estimated.

The SWARM95 experiment had extensive oceanographic measurements, which were carried out almost concurrently with the acoustic measurements.<sup>24</sup> The oceanographic data suggested the presence of solitary internal waves in addition to diffuse internal waves. The solitary internal waves are generated on the shelf edge by interaction of internal tides with the shelf break. These waves occurred on a diurnal cycle. Hence, our analysis was carried over 17 h of data to include the effect of the solitary internal waves. Internal waves are expected to decrease the signal coherence-time. The 400 Hz data indicated a higher standard deviation to the mean ratio than other experimental data (see Table I). The large time variation is presumed to be caused by the internal waves.

#### V. RAGS03 DATA ANALYSIS

The RAGS03 experiments contained two parts: one conducted in September 2003 for underwater acoustic communications research, and the other in the winter of 2003 for array performance study. The data of interest were from the first part of the experiment that took place in the same area as

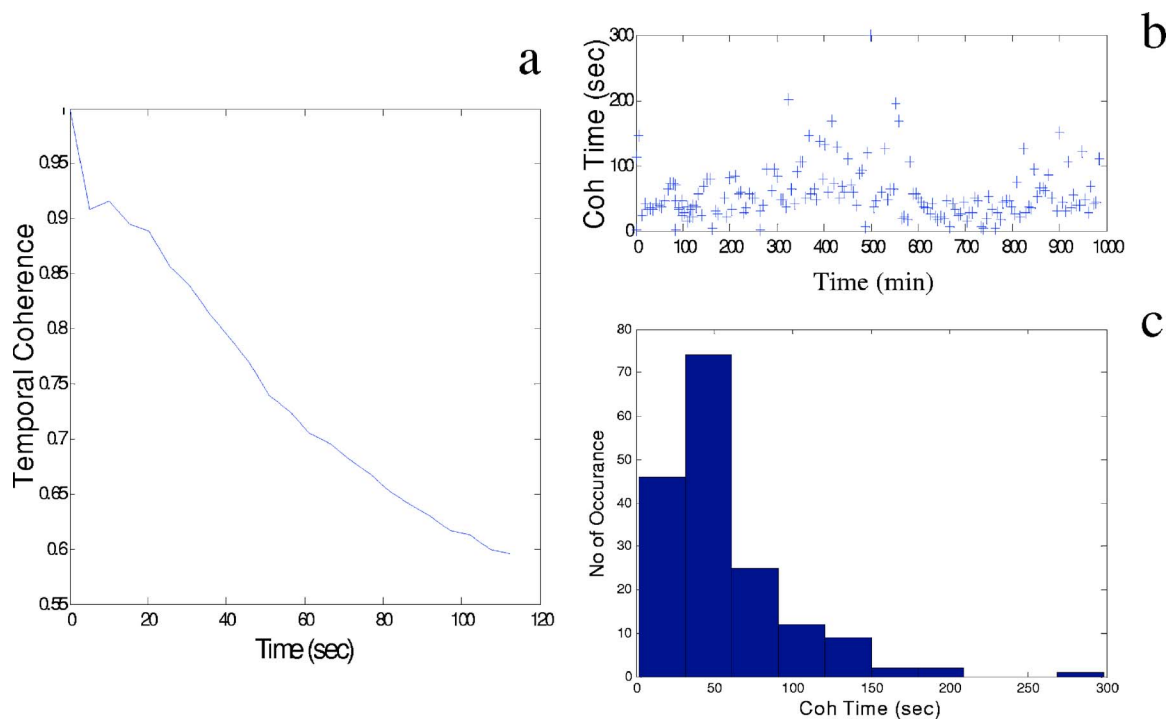


FIG. 17. (Color online) Temporal coherence measured from the  $m$ -sequence data of Fig. 16. Average temporal coherence as a function of signal separation time (a). “Instantaneous” signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c).

SWARM95. The measured sound speed profiles are similar to that found in the SWARM95 experiment; both have the summer profile. There were no concurrent oceanographic measurements with the acoustic measurements as in the SWARM95 experiment, except for CTD casts. Backscattering sonar data reveal the existence of solitary internal waves similar to that found in the SWARM95 experiment.

The source was deployed at a depth of approximately 38 m projecting a broadband signal centered at 20 kHz. The receiver contained 8 hydrophones within an aperture of 2 m, at a mean depth around 41 m. Both were moored from the ocean bottom. The mean water depth is about 70 m. The

source and receivers were separated by 3 km. BPSK signals with a bandwidth of 2 kHz were used to deduce the channel impulse responses. Each packet contained  $\sim 10$  s of data consisting of 19  $m$  sequences each with 1023 symbols.

Figure 19(a) shows the matched filtered output of the  $m$ -sequence data for one packet using the transmitted  $m$ -sequence signal. One observes that the amplitudes of the impulse responses vary significantly between the received  $m$  sequences. Figure 19(b) shows the Doppler shift measured from data of individual  $m$  sequences determined by the peak of the broadband ambiguity function as a function of Doppler frequency and delay time. One sees a substantial varia-

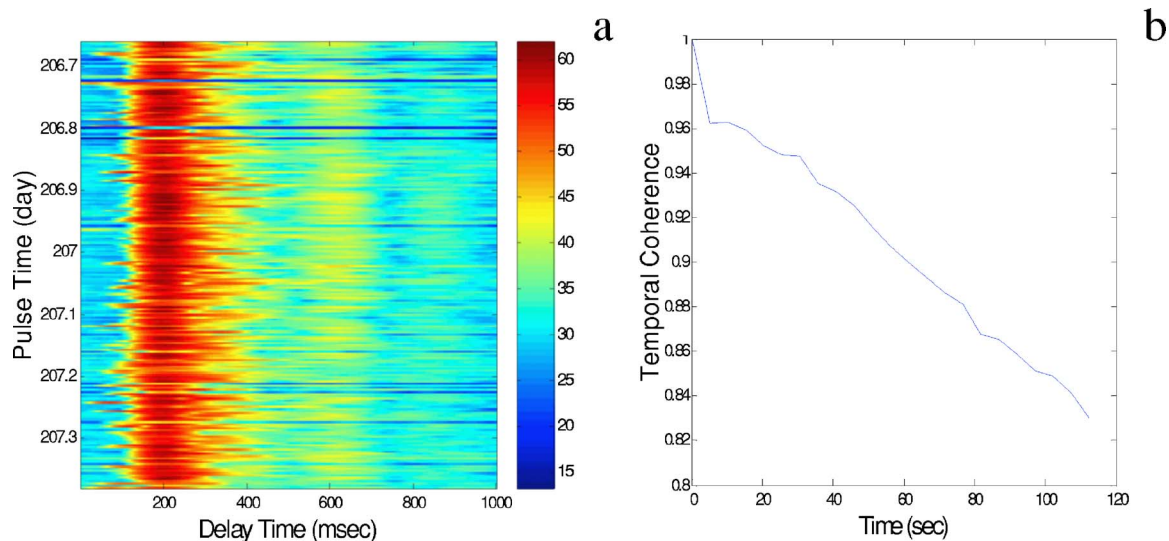


FIG. 18. (Color online) SWARM95 data 224 Hz at a range of 42 km. Channel impulse responses as a function of the delay time and geotime based on  $m$ -sequence signals (a). Multipath arrivals are not resolved due to the limited bandwidth of the signal (16 Hz). Color scale is dB. Average signal coherence as a function of geotime (b).

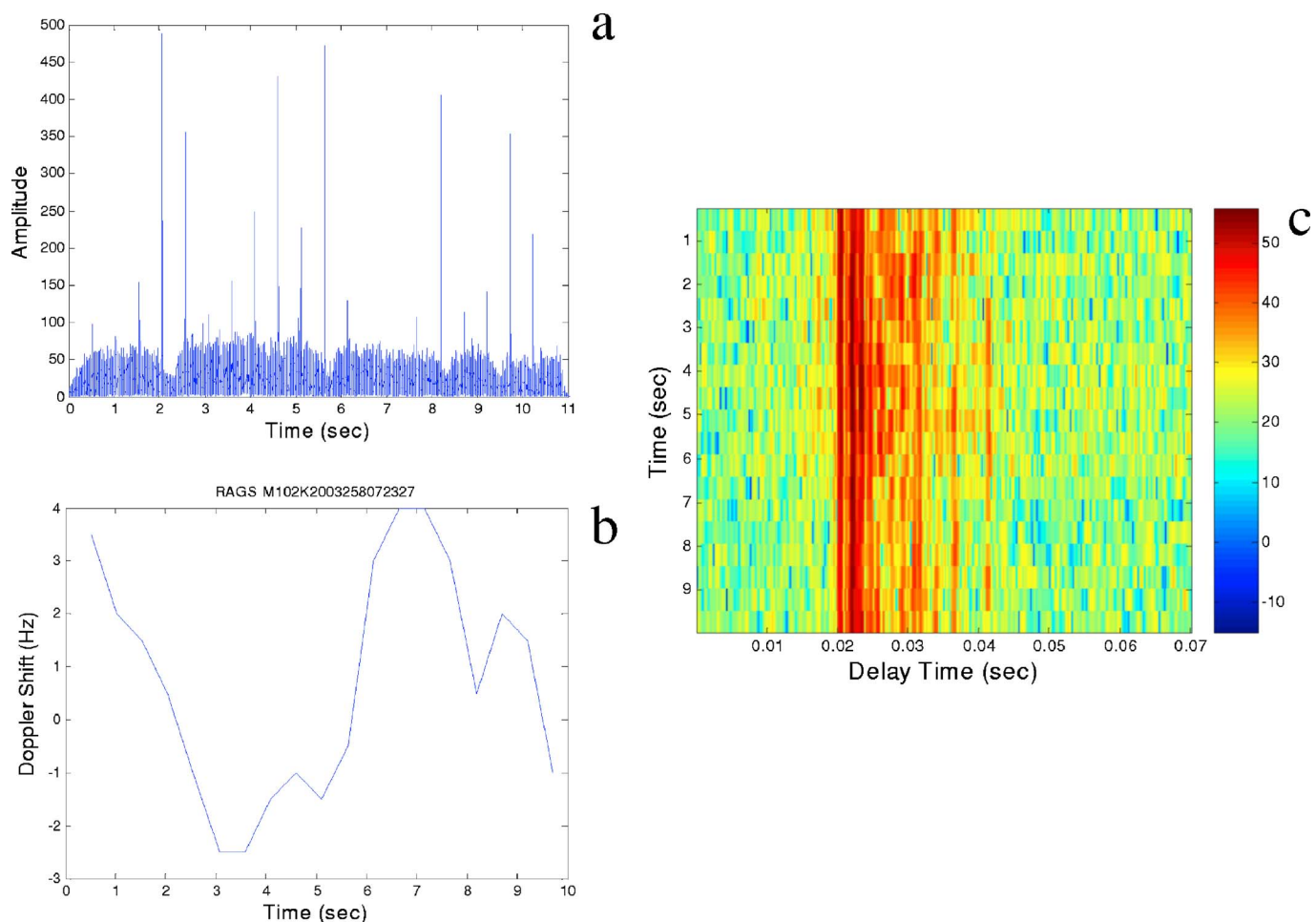


FIG. 19. (Color online) The effect of Doppler shift (array motion) on high frequency (20 kHz) signals. Matched filtered time series of the  $m$ -sequence signals without Doppler correction (a). Doppler shift measured from individual  $m$ -sequence signals using the wide band ambiguity function (b). Impulse response measured from the  $m$ -sequence signals after Doppler correction (c). Color scale is dB. Data were  $m$  sequences from the RAGS03 experiment. Source-receiver range is 3 km.

tion in the Doppler shift, which is nevertheless small compared with the carrier frequency. The small Doppler ratio (Doppler shift over the carrier frequency) indicates a small source/receiver motion which is consistent with the cable/array motion. After applying the Doppler correction, the measured channel impulse responses are plotted in Fig. 19(c). The amplitudes of the Doppler-corrected impulses vary slightly within the packet.

The above analysis shows that Doppler correction is critical for Doppler sensitive waveforms such as the  $m$  sequences, particularly at high (20 kHz) frequencies. Without the Doppler correction, one could underestimate the signal coherence-time. At low frequencies, Doppler correction due to array motion is usually negligible since the Doppler shift is often small compared with the frequency resolution.

To estimate the temporal coherence of the signal, the impulses responses of the  $m$  sequences were processed for each packet of  $m$ -sequence data, as described in Sec. III A. The average of all the temporal coherence curves is plotted in Fig. 20(a), from which one finds a mean coherence-time  $\tau_{0.8} \approx 0.5$  s. Figure 20(b) shows the “instantaneous” coherence-time of individual packets as a function of time. Figure 20(c) shows the histogram of the coherence-time. One finds a mean of 0.5 s and a STD of 0.3 s.

## VI. CHARACTERISTIC PROPERTIES OF THE SIGNAL coherence-time

In this section we investigate the range, frequency, and environmental dependence of the signal coherence-time of the data reported above. For that purpose, the data are characterized in terms of simple empirical fits. The result suggests certain power dependence but such dependence needs, without saying, further testing with more data.

### A. The range dependence

Given the measurement of the signal coherence-time  $\tau_{0.8}$  at two ranges from the ASCOT01 data, one can study the range dependence of the signal coherence-time. The measured (mean) signal coherence-time is plotted in Fig. 21 as a function of range for all three frequencies (450, 1200, and 3550 Hz) in a log-log plot. One finds that the data are well fitted with an inverse square-root power dependence on range [see Eqs. (6) and (7)]. Three lines are used to fit the data, which are given by  $\tau_{0.8} = \delta R^{-1/2}$ , where  $\delta = 2.5 \times 10^3$ ,  $6 \times 10^3$ , and  $3.2 \times 10^4$  for 3550, 1200, and 450 Hz, respectively. The unit of  $\tau$  is s and  $R$  is m.

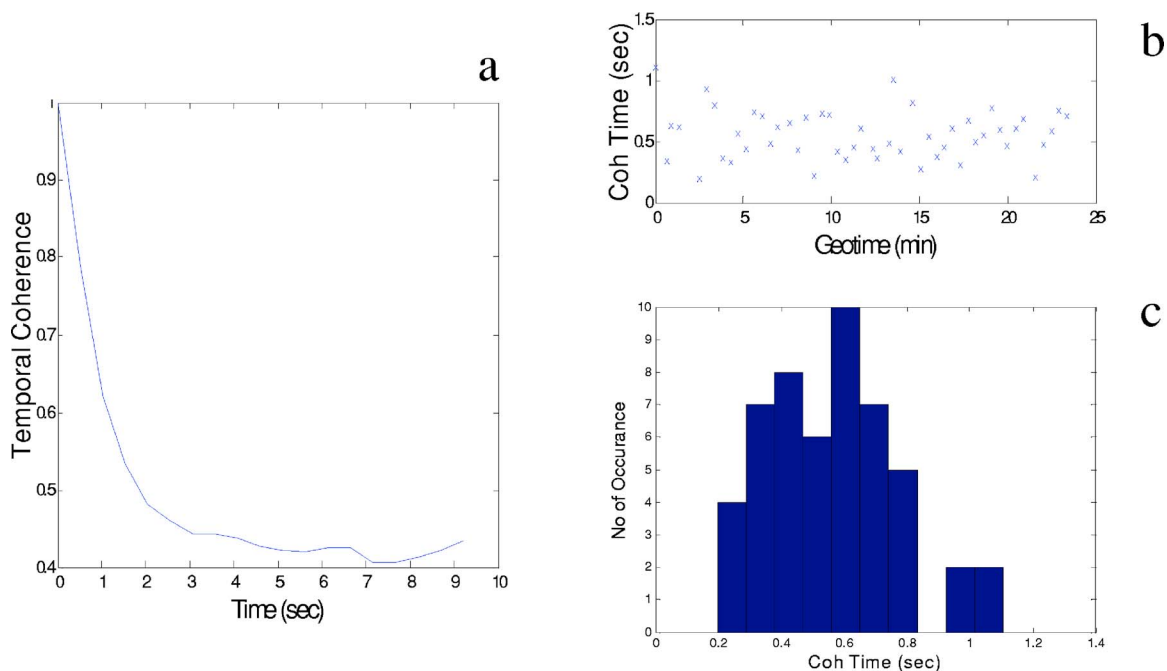


FIG. 20. (Color online) Temporal coherence of high frequency (20 kHz) signals from the RAGS03 experiment. Average temporal coherence of the  $m$ -sequence signals as a function of the signal separation time (a). "Instantaneous" signal coherence-time as a function of geotime (b). Histogram distribution of the signal coherence-time (c).

Deep water data showed that the coherence-time decreases as the inverse square root of the range at long ranges; more precisely, the phase rate increases as a square-root power of range.<sup>25</sup> This range dependence was derived theoretically using the Rytov approximation, as discussed above. At short ranges, the phase rate data in the deep ocean indicate a  $3/2$  power dependence on range for ranges less than 1 km.<sup>25</sup> The above data suggest that the signal coherence-time in shallow water decreases as the square root of range as in deep water. Whether the coherence-time in shallow water at short ranges (also) has a different range dependence is not known.

The ASCOT01 data also contained measurements at 2 km, which could, in principle, be used to address the range dependence. The data are not used in this paper because the data have a high variance and are not conclusive. Note that at a range of 2 km, the signal coherence varies significantly depending on whether the solitary internal waves are within the source-receiver range or not. (At long ranges, one can count on an average number of solitary internal waves in the range.) As such, data over an extended period of time are needed for the temporal coherence measurement to be statistically meaningful. The 2 km range data had a short duration;

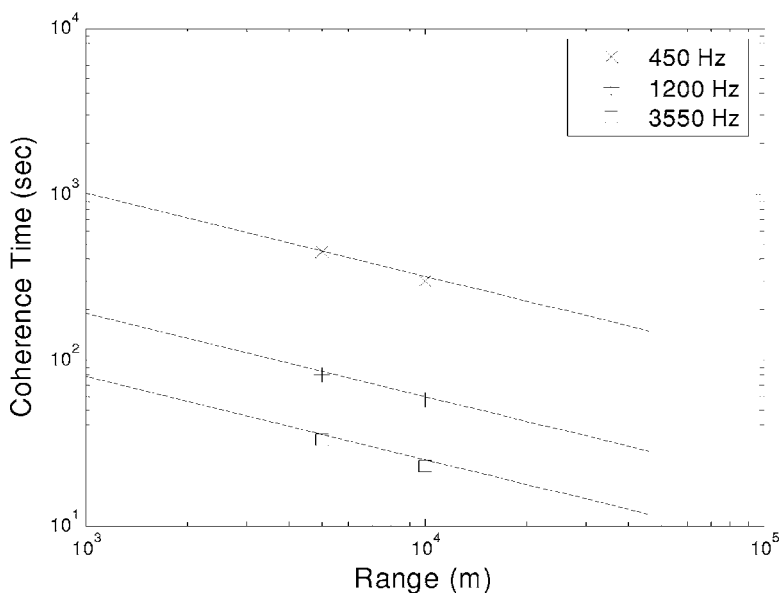


FIG. 21. Signal coherence-time plotted as a function of the range at three different frequencies, based on ASCOT01 data at 5 and 10 km ranges.

TABLE II. coherence-time  $\tau_{0.8}$  and the spread of the coherence-time (in parentheses) measured off the coast of New Jersey; the later reflects the amount of the temporal variation of the coherence-time due to changes in the propagation environment.

| Frequency | $\tau_{0.8}$ at 42 km<br>(SWARM 95) | $\tau_{0.8}$ at 3 km<br>(RAGS03) | $\tau_{0.8}$ at 10 km<br>(Extrapolated) |
|-----------|-------------------------------------|----------------------------------|-----------------------------------------|
| 20 kHz    |                                     | 0.5 ( $\pm 0.3$ ) s              | 0.27 ( $\pm 0.16$ ) sec                 |
| 400 Hz    | 40 ( $\pm 35$ ) s                   |                                  | 82 ( $\pm 72$ ) sec                     |
| 224 Hz    | 130 s                               |                                  | 266 sec                                 |

the short duration may be the reason for the large variance in the 2 km data compared with the mean coherence-time.

## B. Frequency dependence

Based on the theoretical work in deep water [Eq. (7)], the coherence-time [Eq. (6)] is expected to decrease as inverse square-root power of range, and inverse linear power of frequency,

$$\tau_{0.8} = (0.11/\gamma)f^{-1}R^{-1/2}. \quad (11)$$

Assuming the inverse square-root dependence on the range (as studied above), one can study the frequency dependence of the coherence-time  $\tau_{0.8}$ .

The SWARM95 and RAGS03 data were collected at a different range than 10 km (Table II). Using the inverse square-root range dependence, one can extrapolate the coherence-time  $\tau_{0.8}$  to a range of 10 km. The results are shown in Table II, last column. The 10 km data are plotted in Fig. 22. One finds that the data can be fitted with an inverse 3/2 power dependence on frequency, as shown by the curve labeled C.

The coherence-time  $\tau_{0.8}$  measured from the ASCOT01 data at the 10 km range (Table I) is also plotted in Fig. 22. One finds that the data can also be fitted with an inverse 3/2 power dependence on frequency, as shown by the curve labeled B.

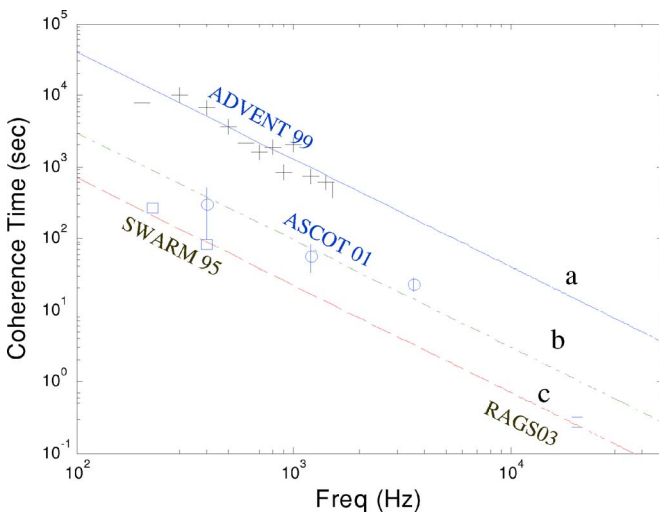


FIG. 22. (Color online) Signal coherence-time plotted as a function of signal frequency based on data from four different experiments. Dashed curves are fit to the data, as described in the text.

In Fig. 22 is also plotted the coherence-time  $\tau_{0.8}$  measured during the ADVENT99 experiment, which was presented in Ref. 13. The temporal coherence was measured using matched field correlation, Eq. (8). The data were narrow band signals (200 to 1200 Hz in steps of 100 Hz) received on a vertical line array. The source receiver range was 10 km. One finds that the data are consistent with an inverse 3/2 power dependence on frequency, as shown by the curve labeled A.

In summary, one obtains an empirical fit of the data presented above as

$$\tau_{0.8} = \beta f^{-3/2} R^{-1/2}, \quad (12)$$

where  $\beta = 4 \times 10^9$ ,  $3 \times 10^8$ , and  $7 \times 10^7$  (m/s) $^{1/2}$  for the ADVENT99, ASCOT01, and SWARM95/RAG03 data, respectively. Using Eq. (6), one finds

$$\nu = \alpha f^{3/2} R^{1/2}, \quad (13)$$

where  $\alpha = 0.11/\beta$ . One thus finds  $\alpha = 0.28 \times 10^{-10}$ ,  $3.6 \times 10^{-10}$ , and  $1.6 \times 10^{-9}$  (m/s) $^{-1/2}$  for the ADVENT99, ASCOT01, and SWARM95/RAG03 experiments, respectively. The 3/2 power dependence on frequency has not been predicted before.

## C. Relation to the sound speed STD

The more random the ocean becomes, the less the signal coherence-time will be. Assuming that the medium fluctuation is the cause for the signal coherence loss, the question is how is the coherence loss related to, or more specifically, how is  $\beta$  (or  $\alpha$ ) related to the oceanographic activities in the water column. This topic obviously requires a detailed modeling analysis and could benefit from some theoretical predictions. For the purpose of data characterization, it is useful to determine the relation of the above coefficient to the sound speed STD, noting that the latter is related to the amplitude of the internal waves and other fine structures.

Let us examine the sound speed STD in the various experiments. The four experiments, ADVENT99, ASCOT01, SWARM95, and RAGS, all took place in a summer environment with a layer of warm water over the cold water. The water depth was approximately 81 m for the ADVENT99 experiment, 100 m for the ASCOT01 experiment, and varied from 57 to 88 m in the SWARM95 experiment. The average sound speed profiles and the sound speed STD can be found in Refs. 13, 22, and 26 for the ADVENT99, ASCOT01, and SWARM95 environments, respectively. The RAGS experiment took place in the same area as SWARM95 and has a similar sound speed profile as SWARM95. The sound speed is nearly constant below 30 m with sound speed increasing slightly (ADVENT99), gradually (ASCOT01), or sharply (SWARM95) to a high sound speed found in the warm layer. The gradual to sharp transitions between the warm and cold layers are responsible for different magnitudes of internal waves, which reside primarily at the pycnocline, which often coincides with the thermocline. The primary difference between the three environments is the sound speed STD, which varies from 0.3 m/s in ADVENT99, to 2–3 m/s in ASCOT01 and 5 m/s in SWARM95, as measured at the depth

TABLE III. Environmental dependence.

|          | Water depth (at receiver) | Thermocline depth | Sound Speed STD ( $\delta C$ ) | Alpha of Eq. (13)      |
|----------|---------------------------|-------------------|--------------------------------|------------------------|
| ADVENT99 | 80 m                      | $\sim 5$ m        | 0.3–0.6 m/s                    | $0.027 \times 10^{-9}$ |
| ASCOT01  | 100 m                     | $\sim 8$ m        | 2–3 m/s                        | $0.28 \times 10^{-9}$  |
| SWARM95  | 88 m                      | $\sim 20$ m       | 5 m/s                          | $1.74 \times 10^{-9}$  |

of the thermocline. (The thermocline is not well defined in the ADVENT99 experiment. Because of the uncertainty of sound speed measurement within 5 m of the surface, the data at 10 m are used for ADVENT99.) The acoustic parameters are summarized in Table III.

The sound speed STD (at the pycnocline) is a good indicator of the strength of the internal waves as it is proportional to the amplitude of the internal waves to the first order of approximation.<sup>19</sup> The Adventure Bank has little internal wave activity. The coasts of New England and New Jersey have medium to high amplitude internal waves. Internal waves create shear instabilities and produce turbulence at the boundaries. These instabilities can affect high frequency sound propagation.

To determine if and how the medium fluctuations affect the signal coherence-time (at a fixed frequency and range), one needs to determine the correlation between the sound speed STD (at the thermocline) and the coefficient  $\alpha$  of the signal phase rate, Eq. (13). We plot in Fig. 23 the value of  $\alpha$  as a function of the sound speed STD as given in Table III. The  $\alpha$  data are consistent with a quadratic power dependence on the sound speed STD as shown by the dashed curve;  $\alpha = 0.6 \times 10^{-10}(\delta C)^2$ .

## D. Observations

The above analysis provides an empirical parameterization of the signal coherence-time with respect to the signal frequency, source receiver range, and sound-speed STD. The result is useful for system design and analysis.

From the physics point of view, the question is how is the signal coherence loss related to the environmental conditions—how to predict the signal coherence loss. As remarked above, there exists (to our knowledge) no theoretical analysis for the signal coherence loss in a time varying shallow water environment. The problem is notably difficult, and is beyond the scope of this paper. Nevertheless, some observations can be made from the data and will be discussed later.

It is generally accepted that the signal coherence-time reflects the time scale of the processes responsible for the signal coherence loss. As such, one can infer from the measured coherence-time the underlying processes causing the signal coherence loss. Signal coherence loss can be caused by signal scattering from the medium heterogeneities, rough surfaces, and bottom inhomogeneities. Which processes are dominant depends on the signal frequency, the magnitude of the inhomogeneities, and the roughness of the boundaries. To properly interpret the data, one needs to model the signal propagation that depends on the sound speed profile and bottom attenuation. The measured channel impulse response should be used to guide and verify the modeling results.

Normally, acoustic rays with shallow depletion/elevation (D/E) angles are trapped in the sound waveguide and thus are most sensitive to medium inhomogeneities such as internal waves and turbulence. On the other hand, acoustic rays with steep D/E angles tend to interact more with surfaces and bottom and are thus more affected by the surface roughness, bottom attenuation, and changes in the overall sound speed. To study the effect of the rough surfaces in a downward refractive sound speed profile, it is preferred to use a source near the surface.

The ASCOT01 experiment used a source near the bottom. As noted above, the late arrivals in the ASCOT01 data (e.g., the 3550 and 1200 Hz signals) change slowly with time. The change in arrival time is, in this case, caused by the change of the layer depth (due to internal tide) rather than the surface height (the external tide) or roughness (wind speed). The reason, as pointed out in Ref. 23, is that the high

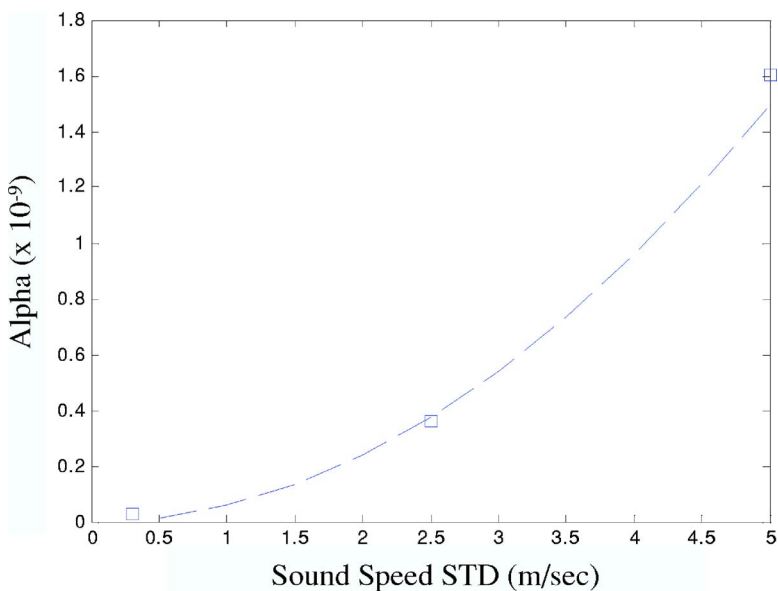


FIG. 23. (Color online) The coefficient  $\alpha$  measured from the data (square) as a function of the sound speed STD.

D/E angle rays have been attenuated, so the signal consists mostly of refracted-bottom-bounced rays. This has also been demonstrated with modeling results in other shallow water environments.<sup>27,28</sup> One finds that the signals from a bottom mounted source, propagated to long ranges, have generally little interaction with the surface in a downward refractive sound speed profile.

Despite the evidence that internal tides affect the multipath arrival time at high frequencies, the internal tides have negligible effect on the signal coherence-time at 3550 Hz in the ASCOT01 environment. This is due to the fact that the signal coherence-time at high frequencies (at the ASCOT01 site) is short (of the order of tens of seconds). This time scale is indicative of the microfine structures (e.g., turbulence). Figure 1 shows rapid fluctuations of the multipath arrivals with time scales on the order of minutes. The results are consistent with the observation that turbulence (and internal waves) has a significant impact on high frequency sound propagation. The instantaneous coherence-time at 3550 Hz [Fig. 4(b)] shows a pattern of random fluctuation, which is expected from random structures like turbulence.

The low frequency (450 Hz) data from the ASCOT01 experiment show, on the other hand, a very stable arrival pattern, particularly for the early arrivals (Fig. 8). (The impulse responses at time 250–270 min are out of alignment, an artifact of data processing.) The instantaneous coherence-times [Fig. 9(b)] are of the same order of magnitude except for certain periods (between 50–100 min in Fig. 8), where the fifth arrival has a high amplitude. Similar observations can be made in Fig. 14 when the third arrival has a high amplitude. At these time periods, the instantaneous coherence-time is higher than other time periods. This is consistent with the fact that a higher coherence-time is obtained when the signal energy is concentrated in a small number of stable multipaths. It is presumed that during these periods, the signals have less scattering and thus have high amplitudes.

The instantaneous coherence-time at 1200 Hz [Fig. 7(b)] shows periodicities on the order of tens of minutes. Similar periodicities are also observed at 450 Hz [Fig. 9(b)]. The origin of this is not known.

The 400 Hz signal from the SWARM95 experiment shows very different properties than the 450 Hz signal from the ASCOT01 experiment, despite the fact that they propagate under similar sound speed profiles. The multipath structures are not as clearly defined in Fig. 16 as in Fig. 8. The multipath arrival times are not as stable in Fig. 16 as in Fig. 8. The reason is that the internal waves off the coast of New Jersey are much stronger than that off New England (See Sec. VI C). Internal waves induce mode couplings in the SWARM95 data,<sup>14,24,27</sup> which smear the arrival pattern as well as temporal stability of the received signals. One notes also that for the 450 Hz signal from the SWARM95 data, the arrival time difference between the early and late arrivals remain approximately unchanged over an 18 h period (Fig. 16), whereas the arrival time difference changes with the tidal cycle in the ASCOT01 data (as discussed above for the 3550 and 1200 Hz signal, a similar pattern was also seen for 450 Hz signal in Fig. 8). The effect of the internal tides on

the signal arrival time is also expected for the SWARM95 signal, but such arrival time differences are easily diminished by the mode coupling effect.

The 224 Hz signal in the SWARM95 data show a dominant group of arrivals with approximately the same total delay time as that seen in the 400 Hz signal. Individual multipath arrivals are not identified due to the narrow band (16 Hz) of the signal. This is not an environmental effect.

## VII. SUMMARY

In this paper, we conducted a systematic measurement of the temporal coherence of acoustic signals propagating in three shallow water areas. The three environments represent areas of low, medium, and high internal wave activity. The measurements cover low (below 1.5 kHz), mid (2 to 5 kHz), and high frequency data. The receiver range covers 3, 5, 10, and 42 km. The “instantaneous” temporal coherence of the signals is also measured, from which one obtained the time variation of the signal coherence-time as well as its mean and STD. To achieve a statistical mean, the temporal coherence is averaged over hours of data, which are generally considered adequate for high frequency signal fluctuations. For low frequency acoustic signals in an area where solitary internal waves are dominant, data beyond 12 h were used to include the solitary internal wave effect. Otherwise, hours of data should be adequate (for diffuse internal waves). An empirical analysis is conducted on the relationship between the signal coherence-time and the acoustic frequency, the source receiver range, and the sound speed STD. It is found that the dependence on frequency and sound speed STD deviate significantly from a previous theoretical prediction for deep water.

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